

Landslides (2026) 23:1935–1950
 DOI 10.1007/s10346-026-02761-6
 Received: 12 September 2025
 Accepted: 21 April 2026
 Published online: 2 May 2026
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 part of Springer Nature 2026

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From clustered landslides to road network disruption: evidence from Typhoon Wipha (2025) in Ningde, Fujian, China

Abstract Typhoon Wipha (No. 6, 2025) triggered clustered shallow landslides in Ningde, Fujian Province, China, in July 2025. A representative failure along National Highway 104 between Fuyang Village and Dashikeng caused severe traffic disruption and required rapid clearance. To characterize pre-failure conditions and mechanisms, we integrated Sentinel-2 image change detection, field mapping, unmanned aerial vehicle (UAV) imagery, digital elevation analysis, meteorological records, rainfall-event comparison, and slope-stability analysis. The representative landslide was small and shallow ($\sim 15,000 \text{ m}^3$) on a vegetated slope (mean gradient $\sim 29.7^\circ$, local relief $\sim 110 \text{ m}$) with a sliding azimuth of $\sim 62^\circ$. Rainfall peaked near the time of failure: 59.4 mm in the preceding 3 h, daily totals of 163.3 and 104.6 mm on 21–22 July (UTC + 8), and a 3-day cumulative of 234.1 mm. Compared with three benchmark typhoon- and hurricane-induced landslide events, Wipha (2025) was characterized by the lowest event-scale rainfall intensity and cumulative rainfall, but the highest antecedent rainfall index ($A_{15} = 22.5 \text{ mm/d}$), and slope-stability analysis indicates that antecedent rainfall played a key preconditioning role in slope destabilization. The coastal bay setting, combined with persistent onshore winds, likely enhanced orographic rainfall and surface loading. Surface soils are classified as sandy loam to loam and comprise kaolinite-rich residuals from granite, favoring transient perched water and hydromechanical softening during intense wetting. Field evidence of windthrow, root–soil cracking, and macropore development indicates a coupled wind–rainfall–vegetation process that promoted preferential infiltration, elevated pore-water pressures, and reduced effective stress, culminating in shallow failure. These findings clarify the controls on typhoon-induced landslides and support corridor-scale susceptibility assessment and resilient road-network planning in typhoon-prone regions.

Keywords Typhoon-induced landslide · Fujian Province · Meteorological Analysis · Landslide characteristics · Catastrophic mechanism · Road network

Introduction

Typhoons pose a major hazard to mainland China, where extreme meteorological forcing frequently triggers severe secondary geohazards (Peduzzi et al. 2012; Shi et al. 2024). Among these, typhoon-induced landslides are typically shallow and spatially clustered on vegetated hillslopes, with well-documented fatalities, prolonged transport disruption, and substantial direct economic losses (Li et al. 2025a, b; Liang et al. 2022; Zhuang et al. 2022). Recent case histories illustrate the scale and diversity of impacts: Fitow (2013) triggered at least 159 landslides and debris flows in Yuyao, Zhejiang,

with direct economic losses of approximately 2.0 billion CNY (Liao et al. 2014). Soudelor (2015) produced more than 200 slope failures in southern Wenzhou, resulting in 12 deaths, four missing persons, and direct losses of 5.014 billion CNY (Guo et al. 2023). Megi (2016) caused catastrophic failures in Zherong, Ningde—three people were reported missing, 207 road collapses occurred, and 66 highways were damaged with 37 interruptions—leading to direct losses of approximately 92.9 million CNY (Yang et al. 2019), and it also triggered a fatal landslide in Su Village, Suichang County that caused 27 deaths on 28 September 2016 (Ouyang et al. 2019; Zhuang et al. 2024). Lekima (2019) generated approximately 2774 rainfall-induced landslides across about 2965 km² in Zhejiang, predominantly small- to medium-sized (Cui et al. 2024a, b), and caused a deadly failure in Yongjia County on 10 August 2019 that resulted in 32 deaths (Cui et al. 2021; Zhuang et al. 2023a, b); it also triggered shallow landslides and debris flows in Daoshi Town, Zhejiang, affecting 25,560 people, with seven deaths and direct losses of approximately 15 billion USD (Liang et al. 2022). More recently, In-Fa (2021) produced landslides that blocked roads and damaged houses in Hongchi Village, Hangzhou (Wang et al. 2022); Doksuri (2023) caused 52 flash-flood and landslide incidents in Fuzhou, affecting approximately 1.455 million people and producing direct losses of 3.053 billion CNY (Yan et al. 2024); and Gaemi (2024) brought rare extreme rainfall to Zixing County, Hunan, triggering widespread clustered landslides that affected approximately 128,000 people, with 50 deaths, 15 missing, and 11,869 houses damaged (Ma et al. 2025; Zhao et al. 2025).

A substantial body of work has identified short-duration, high-intensity rainfall during typhoon passages as the principal regional trigger of landslide clustering (Bessette-Kirton et al. 2019; Chen 2016; Shi et al. 2024; Ye et al. 2020). Taken together, this evidence provides a strong basis for a suite of hazard analysis and risk assessment approaches, including spatiotemporal susceptibility models that incorporate typhoon tracks and forecast-driven numerical predictions of typhoon precipitation fields (Gori et al. 2022; Qin et al. 2024; Wang and Toumi 2022). For secondary hazards, commonly used approaches include landslide susceptibility mapping (Wang et al. 2025); physics-based hazard and risk analyses (Wang et al. 2024; Zeng et al. 2024); typhoon simulators used in resilience engineering (Peduzzi et al. 2012); and assessments of transport-network connectivity and accessibility (He et al. 2025). Nevertheless, the typhoon-driven hazard system is inherently multifactorial and nonlinear: wind, rainfall, slope vegetation, and geotechnical conditions interact through coupled, cascading processes that modulate infiltration, transient pore water pressures, and shear strength, thereby preconditioning and triggering typhoon-induced

landslides. These interactions complicate the prediction of clustered failures, diminish the skill of spatiotemporal risk forecasts, and hinder the design of protection and recovery strategies for road networks in typhoon-prone regions. In particular, the respective roles of antecedent rainfall and event rainfall remain insufficiently differentiated in many typhoon-induced landslide studies, and field-constrained evidence from coastal mountainous southeastern China is still limited, especially for road-adjacent clustered landslides under typhoon wind–rain forcing.

Against this backdrop, Typhoon Wipha (2025) triggered multiple road-adjacent landslides in Ningde, Fujian, providing a representative case for examining failure characteristics and mechanisms in the typhoon-prone coastal mountainous region. Using this case, we investigate the influence of typhoon meteorological forcing and vegetation-mediated hydrological response on landslide initiation. By comparing the rainfall-process metrics derived from this event with those reported for representative typhoon- and hurricane-related landslide cases, we explore both the common and contrasting characteristics of rainfall processes associated with typhoon-induced landslides. To further link these rainfall-process characteristics to regional slope response, we evaluate the respective effects of antecedent rainfall and event rainfall on regional slope stability. Moreover, field investigation indicates that vegetation played an important role during the typhoon process,

emphasizing that typhoon-induced landsliding involves coupled interactions among wind, rainfall, vegetation, terrain, and shallow slope materials. Overall, the results deepen the process-level understanding of typhoon-induced landslide mechanisms and highlight the broader hazard significance of clustered landslides for road-network safety in typhoon-prone coastal mountain regions.

Study area

Ningde City, Fujian Province ($118^{\circ}32'–120^{\circ}43' \text{ E}$, $26^{\circ}18'–27^{\circ}40' \text{ N}$) lies on the southeastern coast of China. The region has a humid subtropical monsoon climate and hilly to mountainous terrain, making it prone to typhoon-induced secondary geohazards, especially landslides (Fig. 1a). In July 2025, Typhoon Wipha (No. 6) produced strong winds and extreme rainfall through its peripheral circulation (Fig. 1b), triggering numerous landslides along multiple highway corridors and severely disrupting traffic in Ningde (Fig. 1c–f). As of 22 July 2025, sixteen large landslides were reported on national and provincial highways in Jiaocheng District—nine on National Highway 237 (G237), three on National Highway 104 (G104), and four on Provincial Highway 306 (S306)—resulting in six road closures (Xiao and Cijian 2025). Specifically, a large landslide occurred on G104 between Fuyang Village and Dashikeng in Feiluan Town (chainage K2213 + 2000, Jiaocheng to Feiluan), adjacent to the G15 Shenhai Expressway (Fig. 1c). Along G237 in

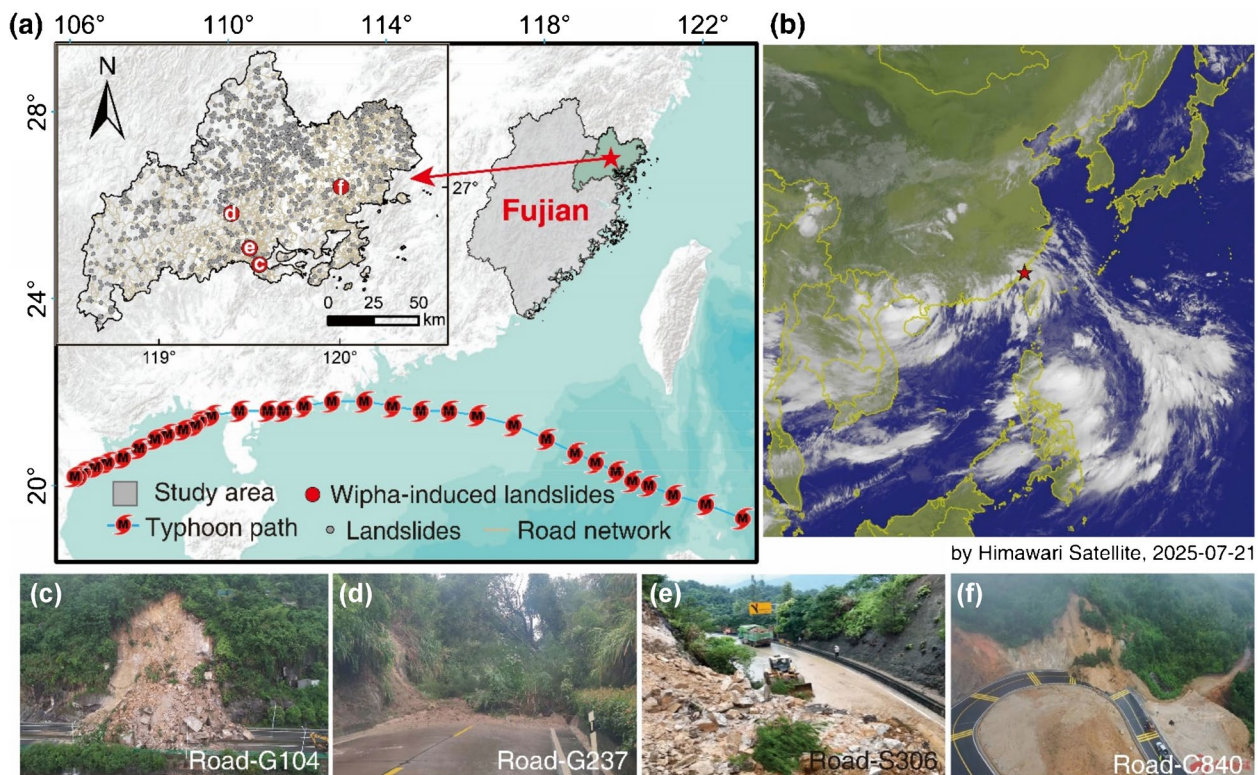


Fig. 1 Overview of the study area and landslide event images, taken in July 2025. **a** The track of typhoon Wipha (2025) and the locations of associated landslides in study area (data sourced from the China Geological Survey). **b** Infrared color cloud map of the study area, data from Himawari meteorological satellite. **c** The photograph of the landslide on National Highway G104 within the study area (obtained from <https://tv.cctv.com/>). **d** The photograph of the landslide on National Highway G237 within the study area (obtained from <https://www.ndwww.cn/xw/ndxw/>). **e** The photograph of the landslide on Provincial Road S306 within the study area (obtained from <https://www.ndwww.cn/xw/ndxw/>). **f** The photograph of the landslide on Road C840 within the study area (obtained from <https://www.cgtn.com/>)

Jiaocheng District, between K1636 and K1654 (Huotong to Pingnan), multiple large slope failures blocked the roadway (Fig. 1d). On S306 at K25 + 650 (Jiaocheng to Gutian), a landslide also blocked the road (Fig. 1e). In addition, a landslide obstructed the C840 road in Changhu Village, Shuimen Township, Xiapu County (Fig. 1f). Collectively, these widespread landslides severely disrupted regional transportation and warrant urgent intervention.

Meteorological background

The Ningde region of Fujian Province is highly exposed to typhoon activity. Rainfall records from the Ningde meteorological station indicate that annual precipitation generally ranges from 1800 to 2800 mm, with rainfall concentrated mainly from June to September (Fig. 2a, b). During 2004–2024, typhoons affecting Fujian occurred recurrently, with a multi-year annual mean of about six events and a clear seasonal peak from July to September (Fig. 2c, d). The close temporal correspondence between intense rainfall and frequent typhoon events indicates that typhoon-related rainfall is a dominant meteorological control on geohazards in the study area.

Geotechnical condition

The geotechnical condition is dominated by weathered granites and Quaternary residual/colluvial deposits (Ding et al. 2023; Xia et al. 2019). The surficial materials consist of a weathered soil matrix with variable proportions of silt, clay, and sandy to silty fines, together with gravel and rock fragments. Slopes typically display a two-layer structure, with loose weathered cover materials overlying competent bedrock. Slip surfaces commonly develop either at the interface between the Quaternary cover and the underlying bedrock or within weakened horizons of the soil mantle (Zhuang et al. 2022).

Under heavy rainfall infiltration, these materials may attain high water contents, reducing shear strength and thereby increasing susceptibility to rainfall-induced instability (Bai et al. 2022).

Vegetation environment

The study area is densely vegetated, with Chinese fir, pine, and bamboo as the dominant vegetation types, together with subordinate shrubs and grasses. In Fujian, typhoon-induced landslides occur predominantly on vegetated slopes with generally high canopy cover (Zhuang et al. 2022). Vegetation modulates both hydrological and mechanical processes, thereby influencing slope stability during typhoon events.

Methods

This study employed an integrated workflow to characterize and diagnose the causes of clustered landslides triggered by Typhoon Wipha (2025) in the study area (Fig. 3). The methodology combines remote sensing interpretation, field investigation, meteorological analysis, and laboratory testing, and further incorporates comparative case analysis and slope stability validation. This framework was used to link landslide characteristics, meteorological background, and geotechnical properties to the diagnosis of cascading failure mechanisms and catastrophic processes.

Multi-source data collection

This study assembles multi-source datasets, comprising remote-sensing imagery, meteorological observations, typhoon best-track records, and digital elevation models (DEM). The specific datasets employed are listed in Table 1, ensuring transparency, traceability, and reproducibility of the results. Typhoon best-track data were

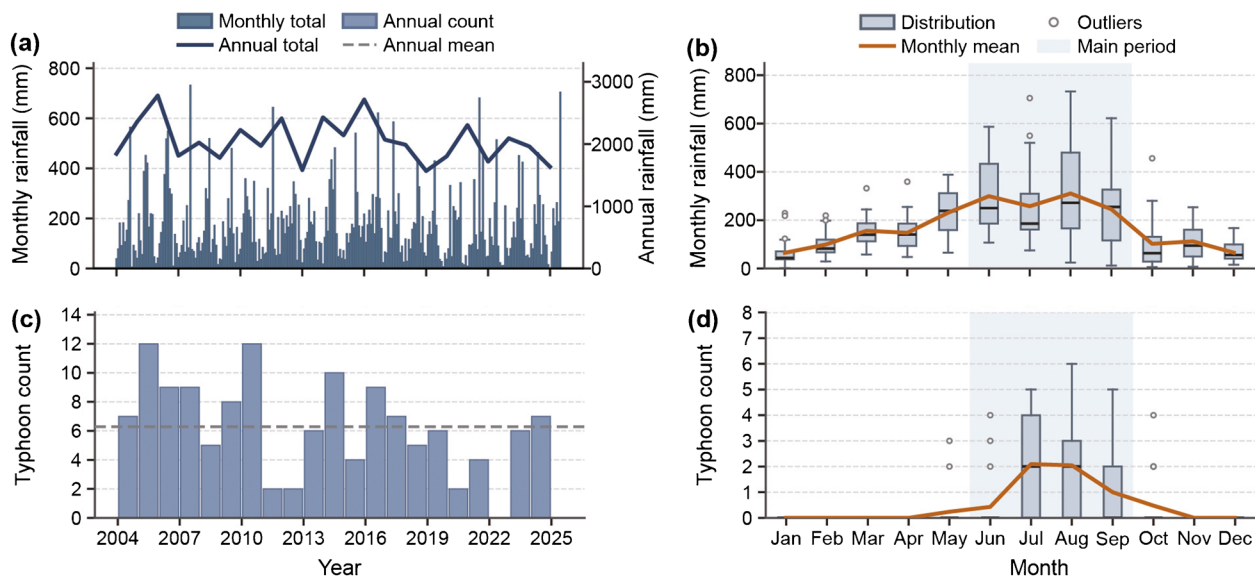


Fig. 2 Rainfall characteristics at Ningde meteorological station and typhoon activity affecting Fujian during 2004–2024: **a** Monthly and annual rainfall totals. **b** Multi-year monthly rainfall distribution at Ningde. **c** Annual number of typhoons affecting Fujian, defined as the deduplicated union of landfalling and severe-impact events. **d** Monthly distribution of typhoons affecting Fujian

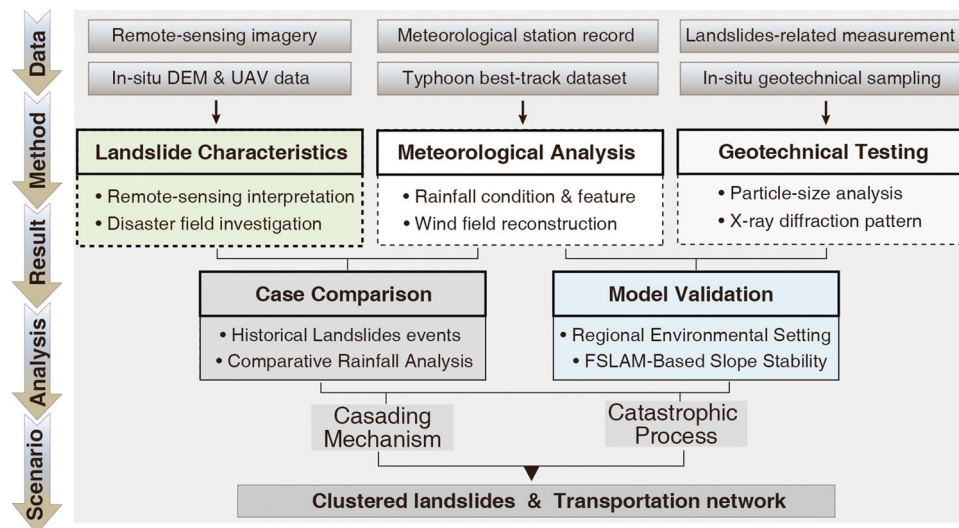


Fig. 3 Methodological framework of this study

Table 1 Data source description

Data type	Description	Source
Sentinel-2 imagery	Google Earth Engine (GEE)	https://dataspace.copernicus.eu/
Google images	Google	https://earth.google.com/
ALOS Digital Elevation Model (DEM)	National Aeronautics and Space Administration (NASA)	https://search.asf.alaska.edu/#/?dataset=ALOS
UAV orthoimage	Field reconnaissance	/
Field photographs	Field investigation and data collection	/
Typhoon best-track data	International Best Track Archive for Climate Stewardship (IBTrACS)	https://www.ncei.noaa.gov/
Meteorological station records	National Meteorological Data Center (CMA)	https://data.cma.cn/

obtained from the International Best Track Archive for Climate Stewardship (IBTrACS), and meteorological station observations were retrieved from the National Meteorological Data Center. Sentinel-2 imagery (Level-2A) was processed on Google Earth Engine (GEE) for atmospheric correction and cloud/shadow masking prior to analysis. The integration of these remote-sensing scenes with event information enabled precise delineation of landslide locations and compilation of per-event attributes (Fig. 4).

Field investigation

The field investigation for this study was conducted as a post-disaster survey, combining unmanned aerial vehicle (UAV) surveys, in-situ mapping, and geotechnical sampling. Specifically, the UAV survey employed a DJI Mavic 4 drone equipped with a network real-time kinematics (NRTK) system to capture high-resolution images and aerial photographs of the landslide location (Fig. 5a).

Subsequent parameter measurements focused on the landslide-affected area, where a detailed survey and mapping analysis of regional conditions and characteristics were conducted (Fig. 5b). To further investigate the causes of the landslide, soil samples were collected at the site (Fig. 5c). These samples were subjected to systematic testing and analysis in the laboratory to examine the influence of soil properties.

Laboratory testing

Laboratory tests were conducted on soil samples collected at the landslide sites to determine particle-size characteristics and mineralogy. Particle-size distribution for the coarse fraction (> 0.075 mm) was obtained using a SYS-31J vibrating sieve, and the fine fraction (< 0.075 mm; No. 200 sieve) was measured with a Malvern laser-diffraction analyser (Fig. 6a). After preliminary grading by dry sieving, the fines were analysed to derive continuous

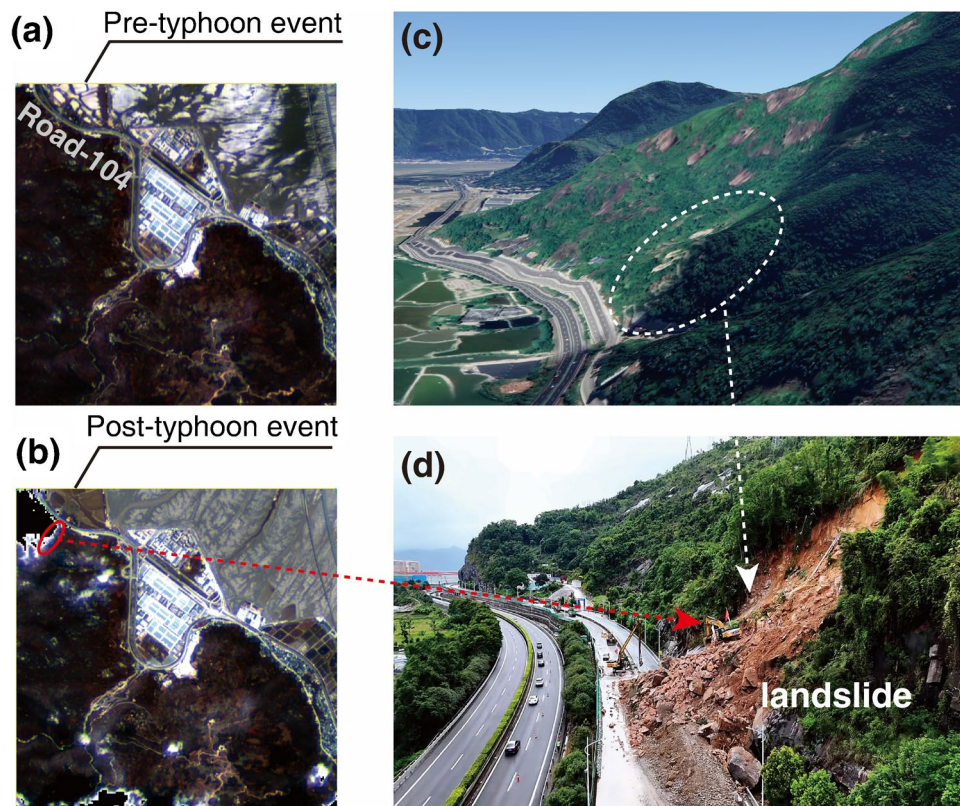


Fig. 4 Multi-source data for landslide identification. **a** Pre-disaster remote sensing imagery (July 2025). **b** Post-disaster remote sensing imagery (August 2025). **c** Google Earth imagery (obtained from Google earth). **d** UAV aerial imagery obtained from eyewitness

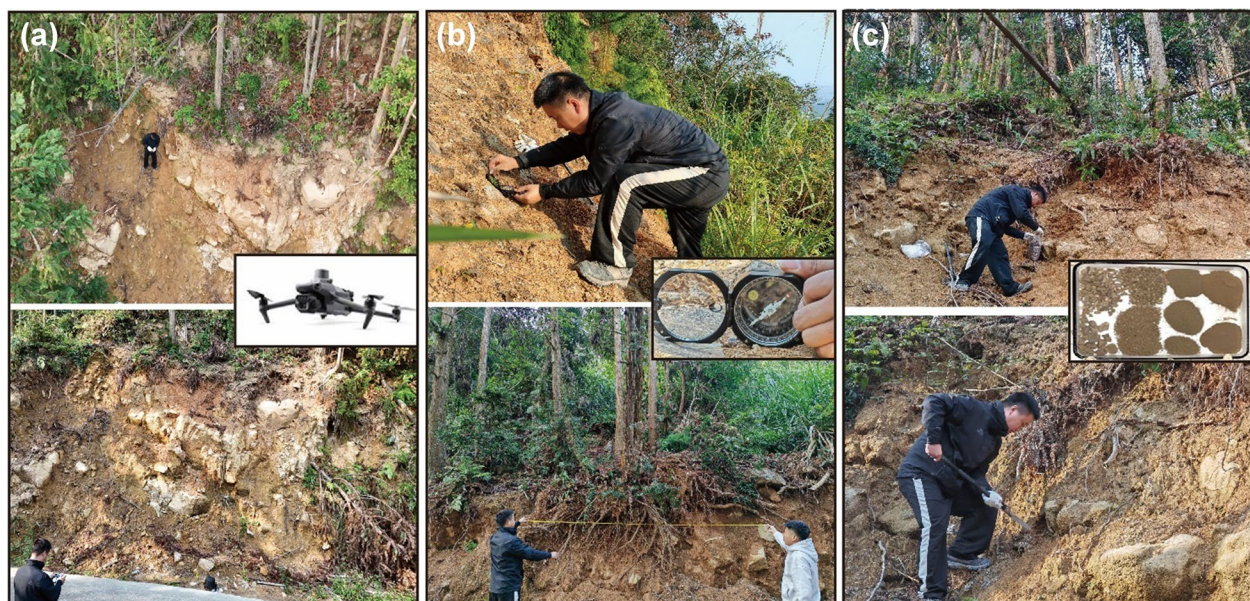


Fig. 5 Field investigation, in-site surveys, and sample collection in the study area. **a** UAV survey. **b** Ground survey. **c** Landslide sample collection



Fig. 6 Sample testing apparatus. **a** Grain size distribution instrument. **b** Mineral composition instrument

particle-size distribution curves for particles < 0.075 mm. Mineralogy was determined by X-ray diffraction (XRD) using a TD-3500 diffractometer (Fig. 6b). Together, these datasets characterize the physicochemical properties of the slope materials and inform interpretation of infiltration behavior and strength evolution relevant to landslide initiation.

Meteorological background analysis

To characterize the meteorological background of the 2025 Ningde landslides, the typhoon parametric wind-field reconstruction based on IBTrACS best-track data was used to identify the main typhoon-affected area over the study region (Supplementary Note 1). Based on the reconstructed influence zone, representative meteorological stations located closest to the core affected area were selected for rainfall analysis. Inter-event comparison was subsequently conducted using the rainfall processes of the 2025 Wipha event and three representative typhoon- or hurricane-induced landslides cases (Fig. S1), namely Morakot (2009), Wipha (2013), and Maria (2017). Gauge records included continuous daily rainfall before and after each event and hourly rainfall during the event (Table S1). Rainfall time series were recorded and analyzed using the regional time reference of each event, with the corresponding UTC offset explicitly indicated.

Antecedent rainfall was represented by index A_{15} , defined as the mean daily rainfall over the 15 days preceding the event rainfall:

$$A_{15} = \frac{1}{15} \sum_{i=1}^{15} P_{d,i} \quad (1)$$

where $P_{d,i}$ is the daily rainfall on the i -th day before the peak-rainfall day.

For inter-event comparison, the event rainfall series was standardized to a common 3-h resolution. Event cumulative rainfall was defined as:

$$E = \sum_{k=1}^N P_{3h,k} \quad (2)$$

Then, mean event rainfall intensity was calculated as:

$$I = \frac{E}{D} \quad (3)$$

where $P_{3h,k}$ is the rainfall amount in the k -th 3-h interval, N is the total number of 3-h intervals within the event, and D is the event duration.

Slope-stability modelling

The antecedent and event-scale rainfall descriptors defined above were subsequently incorporated into the Fast Shallow Landslide Assessment Model (FSLAM) for distributed slope-stability analysis (Cui et al. 2024a, b; Guo et al. 2022; Medina et al. 2021). In this framework, antecedent wetting characterized by A_{15} was used to constrain the antecedent hydrological forcing through the effective recharge term q_a , whereas event cumulative rainfall E was used to define the event-scale rainfall forcing P_e . FSLAM couples the infinite-slope stability framework with a simplified hydrological model combining antecedent lateral recharge and event-scale vertical infiltration. Thus, rainfall influences slope stability through groundwater rise, while slope resistance is controlled by soil shear strength and vegetation reinforcement. Assuming a failure plane parallel to the ground surface, the factor of safety was calculated as:

$$FS = \frac{C_s + C_r}{g \rho_s z \cos \theta \sin \theta} + \left(1 - \frac{(h_a + h_e) \rho_w}{z \rho_s} \right) \frac{\tan \varphi}{\tan \theta} \quad (4)$$

where g is gravitational acceleration; ρ_s and ρ_w are the densities of saturated soil and water, respectively; θ is slope angle; z is soil depth; and C_s , φ and C_r are effective soil cohesion, internal friction angle, and root cohesion, respectively. Here, h_a and h_e denote the groundwater-rise components induced by antecedent recharge and event infiltration, respectively (Supplementary Note 3).

Results

Landslide characteristics

Multiple landslides occurred in Ningde, Fujian Province during and after Typhoon Wipha (2025), underscoring the capacity of typhoons to trigger widespread secondary geohazards in coastal mountainous regions. The failure along National Highway 104 is treated here as a representative case (Fig. 1c). Pre-event and

post-event Sentinel-2 imagery show clear landscape change within the affected area (Fig. 4a, b), although cloud cover associated with the typhoon's outer circulation hindered optical interpretation (Fig. 1b). Consequently, comprehensive field surveys, coupled with UAV imagery, were used to validate and refine the change detection. Ground-based photographs and associated mapping were employed to delineate the landslide boundaries and attributes within the detected change zones (Fig. 4c, d). The landslide is situated in a narrow coastal bay of Ningde City, with an estimated volume of $\sim 15,000 \text{ m}^3$, consistent with a small-scale event. Field investigation, DEM analysis, and UAV data indicate failure on a vegetated slope adjacent to a roadway, with a principal sliding direction (azimuth) of $\sim 62^\circ$. The longitudinal profile shows maximum and minimum elevations of $\sim 132 \text{ m}$ and $\sim 22 \text{ m}$, respectively ($\Delta H \approx 110 \text{ m}$), and a mean slope angle of $\sim 29.7^\circ$ (Fig. 7a, b). The debris comprises a soil–rock mixture with abundant fractured clasts; the slip surface is shallow (on the order of 1–2 m) and locally adjacent to the root–soil interface (Fig. 7c, d). Surface runoff was observed on slopes adjacent to the failure (Fig. 7e). Displaced material covered the carriageway, causing temporary traffic disruption. Because the failure affected a critical transport corridor, emergency clearance operations were initiated immediately, and traffic was restored following debris removal using excavators and loaders (Fig. 7f).

Meteorological analysis

The passage of Typhoon Wipha (2025) and its outer rainbands produced intense rainfall over the Ningde area of Fujian. Precipitation was concentrated during the storm's active phase on 21–22 July 2025, and the landslide was reported at $\sim 19:00$ local time (UTC +8) on 21 July (Lu 2025). Gauge records from the Ningde station indicate 59.4 mm of rainfall in the 3 h preceding failure. Daily precipitation reached 163.3 mm on 21 July and 104.6 mm on 22 July; the cumulative rainfall during the 3 days preceding failure was 234.1 mm (Fig. 8a, b). According to the National Precipitation Grade Standard (GB/T 28592–2012), the rainfall on 21 July falls within the down-pour category. The cumulative rainfall curve steepened markedly during the late stage of 21 July, and the reported failure occurred close to this rapid rise, indicating only limited lag between slope failure and peak storm rainfall. The runoff simulation shows a similarly abrupt increase during the same period (Fig. 8c). Runoff depths were highest in wetland and built-up land, reaching approximately 200–260 mm by the time of failure, whereas forest remained comparatively low throughout the event. Field photographs show that the representative failures occurred on densely forested slopes (Fig. 7c, d), and that visible runoff developed adjacent to the failed slope during the storm (Fig. 7e), providing field

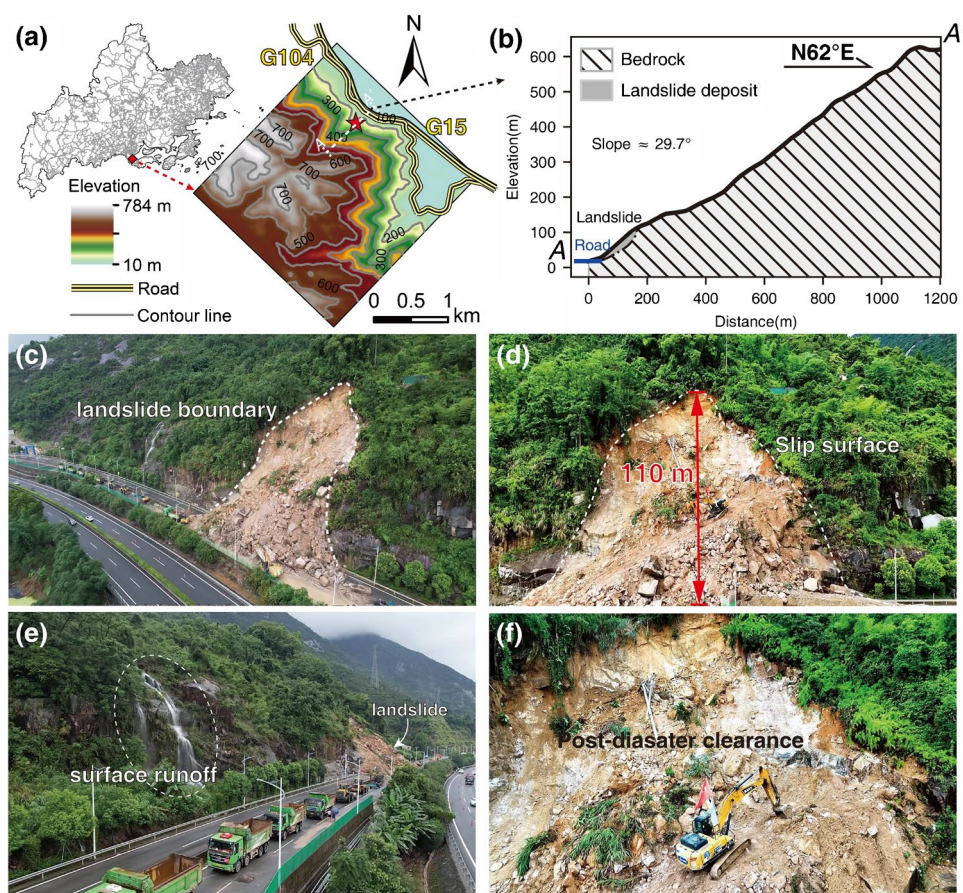


Fig. 7 Geomorphological observations and contextual setting of the 2025 Ningde G104 landslide. **a** Study-area overview showing the landslide location and associated road network. **b** Geological cross-section of the G104 landslide. **c** Field photographs documenting slope failure and surface cracking. **d** Exposure of the slip surface. **e** Oblique overview of the G104 landslide. **f** Post-disaster clearance operations

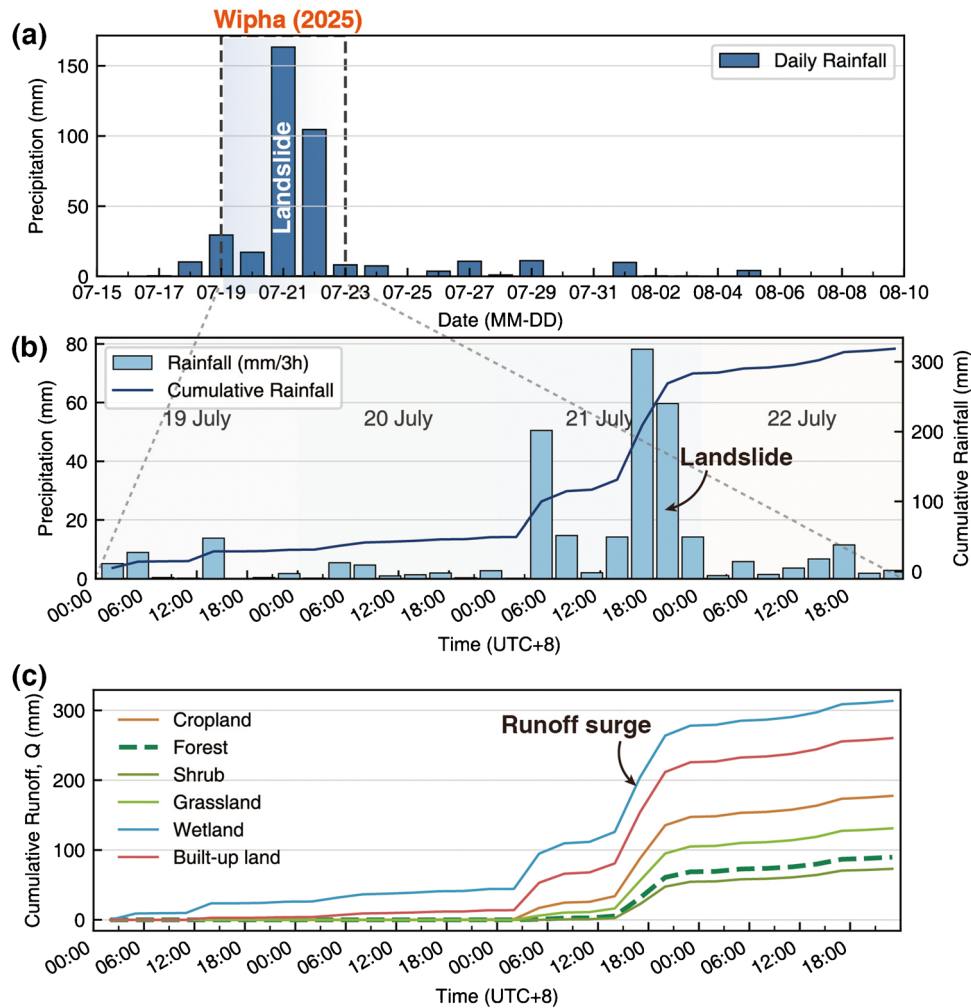


Fig. 8 Rainfall conditions and runoff process in the study area, recorded by Ningde station. **a** Daily rainfall time-series associated with the landslides. **b** Hourly and cumulative rainfall during the typhoon event. **c** Cumulative runoff during the typhoon event

support for the enhanced hydrological response indicated by the modelled runoff series. In this context, the relatively low modeled surface runoff for forest is notable, as it suggests greater rainfall retention and infiltration potential than in other land-cover types. Taken together, the rainfall, runoff, and field evidence indicate that the Ningde landslide developed under strong hydrological loading during the typhoon. Strong winds associated with the typhoon likely further promoted slope destabilization (Buma and Johnson 2015; Lin et al. 2025a, b). The site lies at the mouth of a narrow coastal bay, a setting prone to wind channelling and orographic enhancement of precipitation. Multi-year wind-rose climatology from the Ningde and Xiapu stations shows a persistent onshore (sea-to-land) prevailing wind regime. The slope aspect was north-east ($\sim 62^\circ$), broadly opposing the prevailing onshore winds and consistent with the onshore component (Fig. 9) inferred from a parametric wind-field reconstruction of Typhoon Wipha (2025) based on its track (Supplementary Note 1). Although the site was not immediately adjacent to the storm's center, windward exposure under the peripheral circulation likely promoted slope destabilization during the event.

Event-based rainfall comparison

Comparison of the event-based rainfall metrics reveals clear contrasts among the four benchmark cases of typhoon- and hurricane-induced landslides (Fig. 10). Morakot (2009) and Wipha (2025) show relatively long durations ($D = 87.0$ and 102.0 h, respectively), whereas Wipha (2013) is characterized by the shortest duration ($D = 24.0$ h) and the highest mean rainfall intensity ($I = 34.33$ mm/h). Maria (2017) has an intermediate duration ($D = 66.0$ h) but relatively high cumulative event rainfall ($E = 594.4$ mm). Among the four cases, Wipha (2025) has the lowest mean event intensity ($I = 3.16$ mm/h) and the lowest cumulative event rainfall ($E = 322.6$ mm), but the highest antecedent rainfall index, with $A_{15} = 22.5$ mm/d. Its rainfall series also shows a pronounced antecedent rainfall phase prior to the main event window, which is less evident in the other three cases. The intensity-duration comparison further separates the Ningde case from the other benchmark events (Table S1). Morakot (2009), Wipha (2013), and Maria (2017) all plot above their corresponding empirical I - D threshold curves, whereas Wipha (2025) plots below the Fujian threshold curve (Fig. 11). In

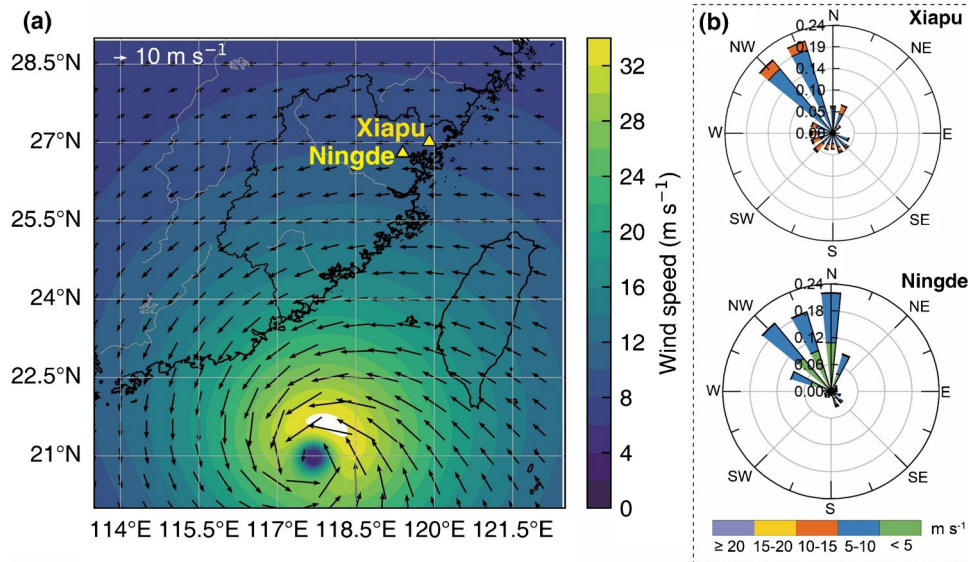


Fig. 9 Spatial distribution and characteristics of wind field in the study area. **a** Wind field during typhoon passage based on a parametric typhoon wind-field model. **b** Twenty-year wind roses at Ningde and the nearby Xiapu station

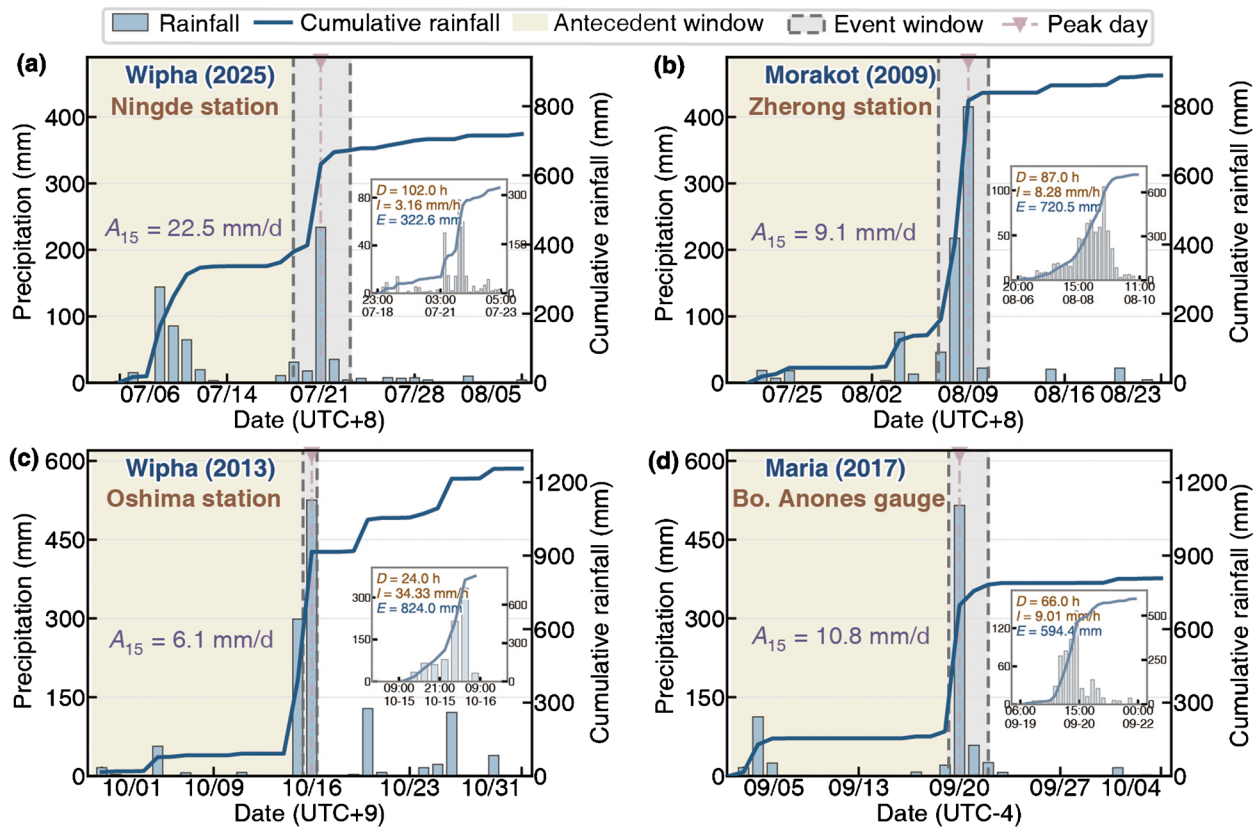


Fig. 10 Event-based rainfall characteristics of four typhoon- and hurricane-induced clustered landslide cases. **a** Wipha (2025), Ningde station, Fujian, China. **b** Morakot (2009), Zherong station, Fujian, China. **c** Wipha (2013), Oshima station, Japan. **d** Maria (2017), Bo. Anones gauge, Puerto Rico, USA

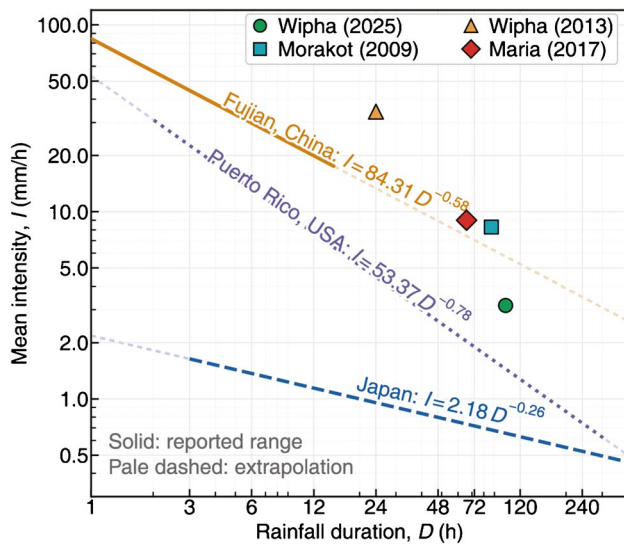


Fig. 11 Comparison of rainfall intensity–duration thresholds and four representative typhoon/hurricane-induced landslide events

addition, although Morakot (2009) and Wipha (2025) both affected the Ningde area and share similarly long rainfall durations, they differ markedly in event-scale rainfall magnitude and threshold position. Overall, these results indicate that the Ningde landslide case was not controlled solely by event rainfall during the main typhoon period, but was also associated with a pronounced antecedent rainfall process. This suggests that antecedent rainfall, by conditioning the wetness state of the slope soil, is an important factor affecting regional slope stability.

Regional soil properties

We conducted particle-size analysis on surface soils collected at the landslide site. The mass-based fractions for the full sample set were 0.8–3.9% gravel, 49.3–64.6% sand, 25.7–40.9% silt, and 5.8–7.6% clay (Fig. 12a). After normalising to the fine-earth fraction (< 2 mm), the samples plot as sandy loam (HP1, HP3, HP4) and loam (HP2) on the USDA textural triangle (Soil Science Division

Staff 2017). HP1 and HP4 plot near the loam boundary, whereas HP3 plots near the sandy-loam boundary (Fig. S3). The fine-grained content (silt and clay) spans ~31.5–48.5%, indicating moderate to relatively high fines. X-ray diffraction (XRD) patterns show peaks attributable to kaolinite, quartz, orthoclase, and muscovite. Semi-quantitative peak-area proportions are ~40.7% kaolinite, ~23.9% quartz, ~22.2% orthoclase, and ~13.2% muscovite (Fig. 12b). Thus, kaolinite is the most abundant phase overall—and the dominant clay mineral—while quartz and orthoclase are the principal non-clay constituents. This assemblage is characteristic of residual soils derived from granitic parent material in southeastern coastal China (Lan et al. 2025). The observed textural and mineralogical attributes indicate that the relatively high fines content is associated with relatively low hydraulic conductivity, thereby favoring the formation of transient perched water tables during intense or prolonged rainfall (Bai et al. 2022). These regional soil characteristics therefore indicate a pronounced sensitivity to rainfall infiltration. Under the combined effects of antecedent and event rainfall infiltration, the reduction in soil strength is likely to be more marked, highlighting the need to explicitly consider hydrologic infiltration in interpreting slope response and instability. During typhoon rainfall, infiltration raises pore-water pressures and reduces matric suction; concurrent wetting induces fabric loosening and hydro-mechanical softening, thereby reducing effective stress and shear strength. Although kaolinite has limited swelling potential relative to smectite-bearing soils, saturation-driven softening and micro-structural dispersion can still decrease shear resistance, increasing susceptibility to deformation and shallow failure during extreme rainfall events.

FSLAM-based stability analysis

The FSLAM simulations show a progressive but strongly non-linear reduction in slope stability from the original condition to the event-rainfall condition and, more distinctly, to the antecedent-plus-event rainfall condition (Fig. 13). Under the original-state simulation, stable cells dominated the study area, whereas low-FoS zones were sparse, localized, and spatially discontinuous. Under event rainfall alone, the regional FoS pattern changed only slightly, and most slopes remained above the critical threshold of FoS = 1. By contrast, after antecedent rainfall

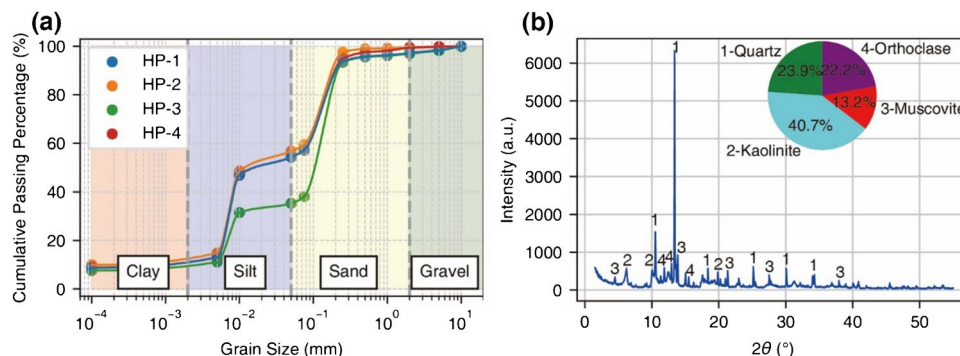


Fig. 12 Properties and characteristics of soil samples from Ningde region. **a** Particle size distribution curve of the soil. **b** X-ray diffraction analysis of the soil sample

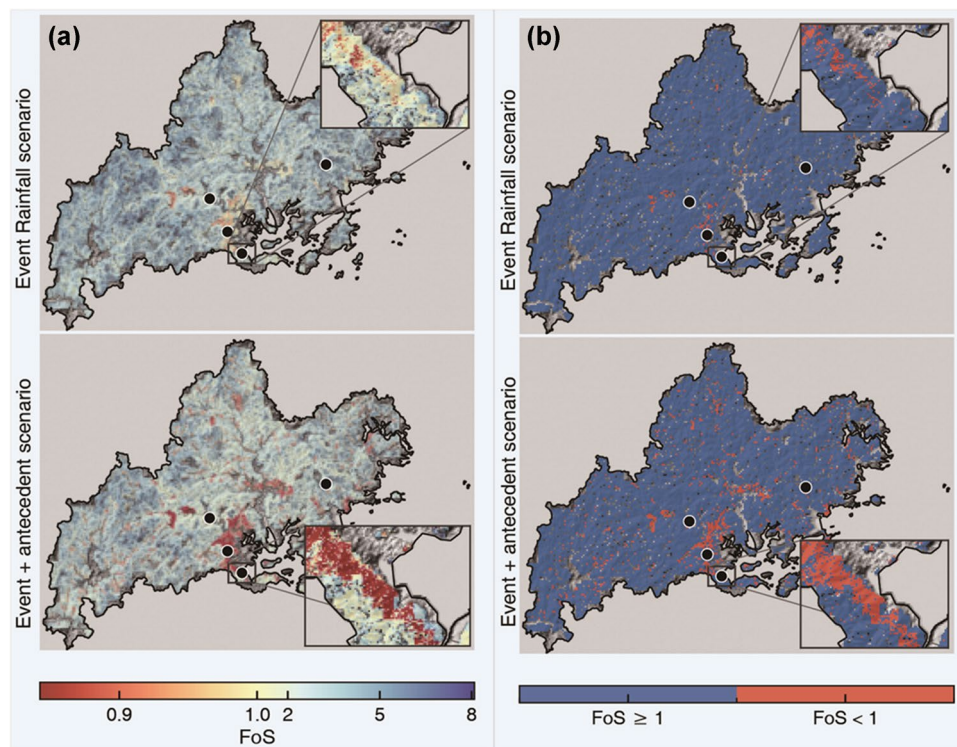


Fig. 13 Factor of safety (FoS) and slope instability classification under different rainfall scenarios. **a** Factor of safety (FoS). **b** FoS binary classification

was incorporated, low-FoS cells expanded markedly and became more spatially aggregated, forming broader and more continuous belts, particularly in the road-adjacent mountainous sector and within the enlarged landslide-prone area. The local zoom-in views show the same pattern: areas characterized by scattered low-FoS pixels under the original and event-rainfall conditions evolved into wider and more coherent unstable belts under the combined-rainfall condition.

The point-based results are consistent with the map-based patterns (Fig. 14). At the four representative slopes (P1–P4), FoS values decreased from 1.347, 1.083, 1.313, and 1.440 under the original condition to 1.205, 1.060, 1.268, and 1.392 under event rainfall alone, indicating only limited reductions and no threshold exceedance. After antecedent rainfall was added, FoS further decreased to 0.550, 0.669, 1.029, and 0.785, respectively. Consequently, P1, P2, and P4 became unstable, whereas P3 remained marginally stable, only 0.029 above the critical threshold. The mean FoS of the four representative slopes declined from 1.296 in the original state to 1.231 under event rainfall and then to 0.758 under the antecedent-plus-event rainfall condition. Correspondingly, the number of unstable representative slopes increased from 0/4 under the original and event-rainfall scenarios to 3/4 under the combined-rainfall scenario. Overall, the simulations indicate that the principal stability reduction occurred only when antecedent wetting was considered, whereas the destabilizing effect of event rainfall alone remained comparatively limited.

Field failure mechanism

Field investigations show dense vegetation cover, and landslides predominantly occurred on forested (tree-covered) slopes, where windthrow and uprooted trees were common (Fig. 15a). Post-typhoon, separation cracks formed at the root–soil interface (Fig. 15b), and soil adjacent to tree roots on slope surfaces was loosened (Fig. 15c); distinct cracks and holes were also observed (Fig. 15d). These features are likely attributable to wind force during typhoon: wind loading on the canopy was transmitted to the root–soil system, mobilizing and bending roots and disrupting the surrounding soil structure, thereby promoting the development of macropores and preferential flow paths. Field photographs also document surface runoff, indicating that intense rainfall enhanced the near-surface hydrologic response (Fig. 7e). Rainwater flowed downslope and rapidly infiltrated via preferential pathways along cracks and macropores around root networks (Lin et al. 2025a, b). Such preferential, nonuniform infiltration recharged shallow groundwater within the hillslope, preconditioning the slope to failure (Qin et al. 2022). Sustained infiltration and moisture redistribution can focus pore-water pressure within the hillslope, generating high-pressure zones that reduce effective stress and weaken shear resistance. With continued rainfall, progressive saturation elevates water content and decreases shear strength, increasing the likelihood of failure and ultimately triggering landsliding. These observations corroborate

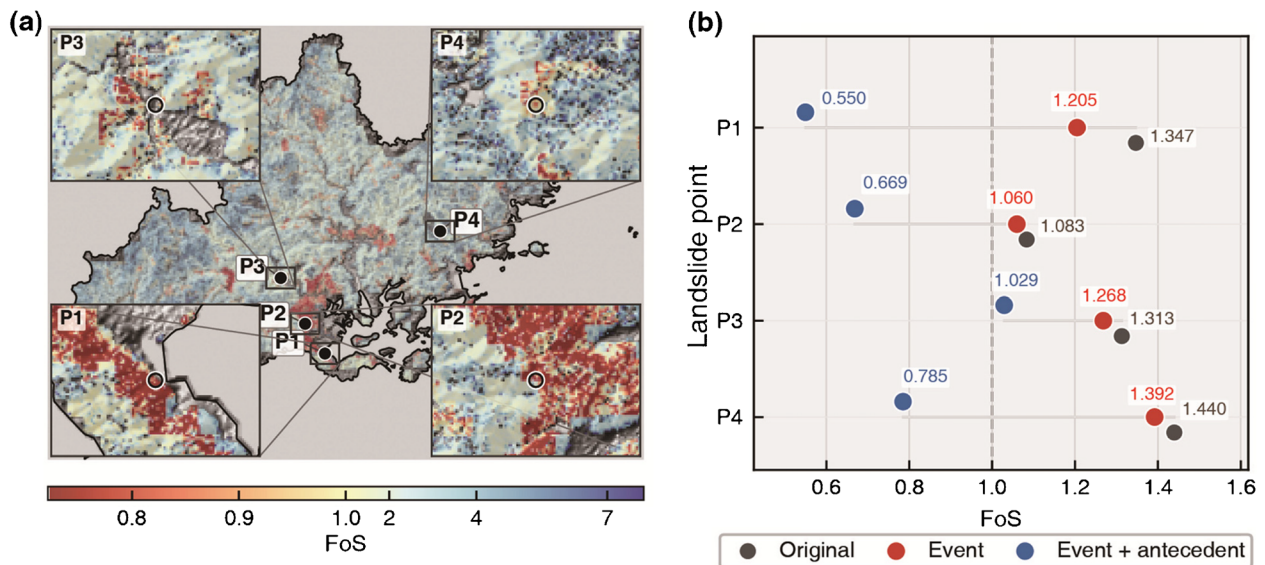


Fig. 14 Comparison of FoS maps and point-based FoS under different rainfall scenarios. **a** Spatial distribution of FoS with local zoom-in views. **b** Comparison of point-based FoS values

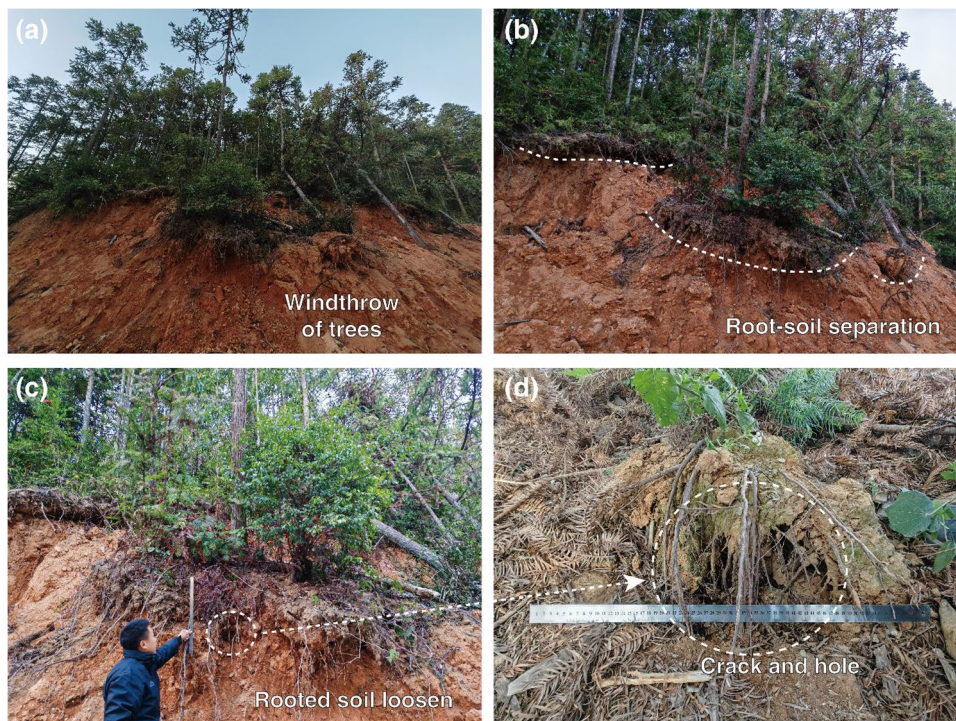


Fig. 15 In situ field evidence of landslide-related features. **a** Post-typhoon tree uprooting and windthrow. **b** Root-soil separation. **c** Loosening of rooted soils. **d** Root-adjacent cracks and voids

prior findings on typhoon-induced landslides (Zhuang et al. 2022; Zhuang et al. 2023a, b), refine existing interpretations, and provide additional field evidence. Collectively, the hydromechanical coupling of strong winds and intense rainfall acting on forested

slopes—initiating a tree-induced cracking-percolation cascade that culminates in seepage-driven failure—provides a plausible mechanism for deformation and failure in typhoon-affected coastal hillslopes.

Catastrophic mechanism of typhoon-induced landslides

Field observations indicate that typhoon-induced landslides on vegetated slopes are controlled by coupled wind–rainfall–vegetation interactions rather than by rainfall forcing alone. Cascading coupling among typhoon rainfall processes and other conditioning factors underlies the complex failure mechanisms of these landslides. The clustered failures triggered by Wipha (2025) in Ningde illustrate this multi-factor cascading process and its associated failure mechanism (Fig. 16). Previous studies have likewise shown that failures under typhoon conditions arise from the co-action of wind loading, high-intensity rainfall, and vegetation disturbance (Zhuang et al. 2022). The co-occurrence of strong wind and torrential rainfall constitutes the primary atmospheric forcing, and the contribution of wind should not be ignored in typhoon-induced landslides (Lin et al. 2025a, b; Parra et al. 2021). Wind loading disturbs forests and overturns trees (Buma and Johnson 2015; Feng et al. 2020), opening cracks and voids adjacent to the root–soil matrix that act as preferential flow pathways for rapid infiltration (Guo et al. 2024; Hu et al. 2024a, b). Such disturbance loosens the near-surface soil fabric and reduces the root-reinforced strength of the soil mantle, preconditioning slopes for rainfall-triggered failure (Sun et al. 2021). The resulting preferential flow increases saturation, reduces matric suction and effective stress, and thereby weakens shear strength (Bessette-Kirton et al. 2020; Zhuang et al. 2023a, b), consistent with rainfall-induced landslides during typhoon passages (Liang et al. 2022; Ma et al. 2025; Wang et al. 2022; Zhao et al. 2025). Beyond these wind–rainfall–vegetation interactions, antecedent rainfall was also a key preconditioning factor in the typhoon-induced landslides.

Both the rainfall-process comparison and the stability analysis suggest that substantial wetting had already occurred before the most intense event rainfall, thereby increasing background moisture, promoting groundwater response, and lowering the threshold for failure. This effect is particularly important in granitic residual soils of southeastern coastal China, where infiltration and prolonged wetting can soften and disintegrate the soil fabric, leading to marked hydro-mechanical weakening and strength reduction (Ding et al. 2023; Wei et al. 2019). This weakening may be further exacerbated by root-related hydrological effects, which enhance water penetration and further reduce the inherent strength of the slope material (Hou et al. 2024; Mao et al. 2024). Field evidence nevertheless indicates that event rainfall remained essential for final failure initiation. Extreme rainfall can also generate surface runoff that raises basal shear stresses and increases soil mobility (Li et al. 2025a, b; Xia et al. 2019). The interaction between vegetation and hillslope hydrology—through preferential flow, subsurface stormflow, and runoff-driven erosion—plays a central role in promoting instability (Fan et al. 2015; Hu et al. 2024a, b; Jiang et al. 2023). Infiltrating water may propagate along cracks and root channels to depth, locally inducing piping and void enlargement and weakening the soil matrix until failure occurs (Zhang et al. 2019). Taken together, these processes, coupled with antecedent hydrological conditioning, contributed to the clustered typhoon-induced landslides in Ningde. Reliable prediction of typhoon-induced clustered landslides therefore requires joint consideration of antecedent rainfall, event rainfall, and wind-induced vegetation disturbance. Neglecting antecedent wetting, or over-simplifying event-rainfall infiltration under disturbed vegetation cover, will limit the ability of physically based models to reproduce real typhoon-driven hillslope responses.

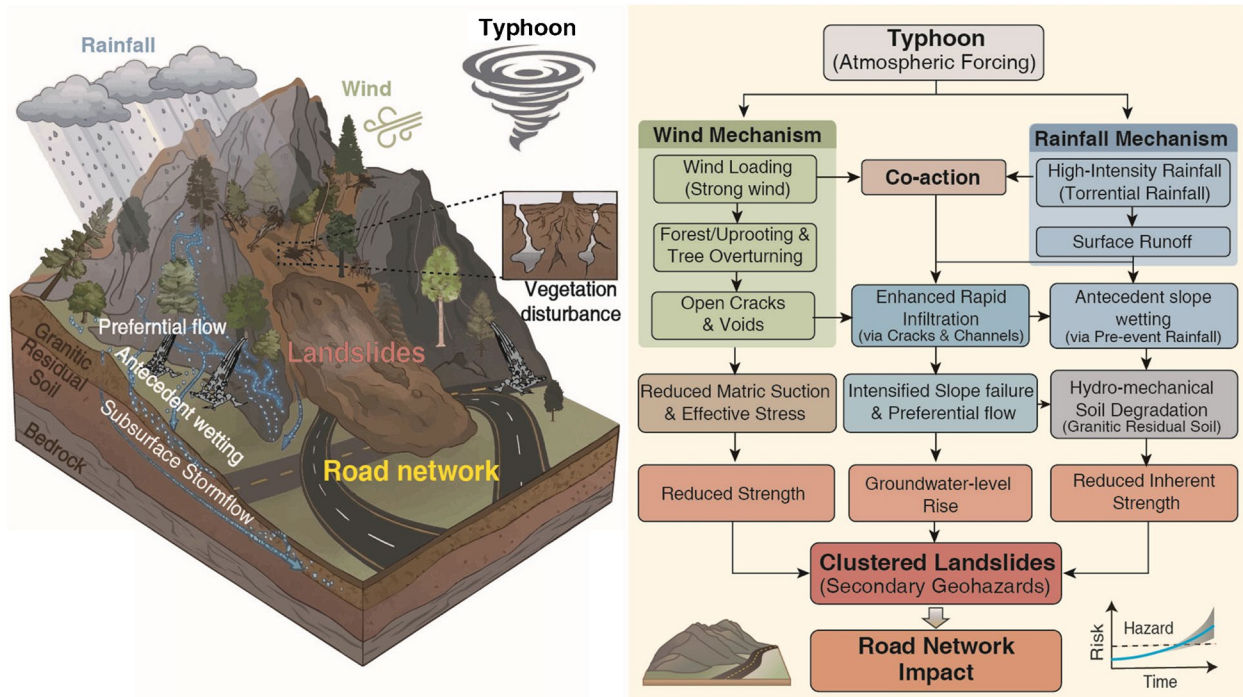


Fig. 16 Illustration of typhoon-induced clustered landslides and coupling effect for landslide processes within the road network

Impact of landslide clusters on transportation

Typhoon-induced landslide clusters and their associated transportation risk deserve particular attention, because clustered slope failures during typhoon events can obstruct road traffic and severely disrupt the regional road network. In this context, spatial prediction of landslide risk under typhoon scenarios is critical for disaster prevention, mitigation, and emergency traffic management. The persistent and cumulative disturbances induced by typhoons significantly exacerbate the vulnerability of road networks, as evidenced during Typhoon Wipha (2025)—which triggered multiple landslides and severe disruptions across Ningde's regional road network. Within mountainous corridors susceptible to typhoon impacts, landslide clusters can obstruct roadways and reduce traffic capacity; some sections experience complete closures, thereby degrading regional network connectivity. This pattern has been documented elsewhere (Liu et al. 2012; Ramos-Scharrón et al. 2021; Xiong et al. 2021). Failure to promptly clear debris and restore service after clustered-landslide road blockages impairs regional road capacity and network performance. These impacts underscore the need for transport authorities to implement typhoon-specific management strategies (Fujisawa et al. 2010; Miao et al. 2018). A comprehensive warning-and-response system is essential to manage typhoon-induced landslides and their impacts on road networks. Pre-typhoon measures should include: (i) assessing landslide susceptibility and corridor safety along slope-adjacent road segments (Lee et al. 2018; Tseng et al. 2018; Wang et al. 2025); (ii) pre-identifying critical sections and planning redundant detours and evacuation corridors (Fahad et al. 2019; Feng and Lin 2022; Wang and Hu 2024). During typhoon impacts, transport authorities, meteorological services, and disaster emergency management departments should jointly monitor hydrometeorological conditions and road status in real time to inform phased closures, traffic control measures, and prioritized obstacle clearance. Collectively, proactive risk assessment, infrastructure reinforcement, and adaptive traffic management are key to mitigating the impacts of typhoon-induced landslides on transportation and enhancing the resilience of mountainous trunk roads.

Conclusions

Typhoon-induced landslides represent a severe problem in the mountainous regions of China's southeastern coast. The clustered landslides triggered by Typhoon Wipha (2025) in Ningde, Fujian, underscore the risks associated with secondary geological disasters caused by typhoons. This study reveals the complex mechanisms behind these landslides, which were caused by the combined effects of strong winds, torrential rainfall, and vegetation interactions. It further shows that antecedent rainfall and event rainfall acted jointly in the failure process, with antecedent rainfall providing hydrological preconditioning and event rainfall contributing to final failure initiation. Furthermore, the landslides had significant repercussions on the road transportation system, particularly in vegetated mountainous slopes, where road accessibility was severely compromised. To better address the challenges posed by typhoon-induced landslides, it is essential to strengthen disaster preparedness, post-disaster recovery efforts, and transportation planning, ensuring the continuity and resilience of the transportation system.

The draft manuscript contains supplementary material.

Funding

This research was funded by the National Key R&D Program of China (2024YFC3012701), the Science and Technology Research Program of the Institute of Mountain Hazards and Environment, Chinese Academy of Sciences (Grant No. IMHE-ZDRW-07), and the Social Survey Project of the University of Chinese Academy of Sciences.

Data availability

All data supporting the findings of this study are from the corresponding author upon request.

Code availability

Data analysis and processing were conducted using the software python. The codes and scripts used for this study are available from the corresponding author upon request.

Declarations

Competing interests The authors declare no competing interests.

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Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10346-026-02761-6>.

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